LOOK AFTER egiver

G-OLD x UAiTED
University Alliance in Talent Education Development







Problems

Costly Homecare Services and Products



Shortage of Healthcare Workers

TAIPEI TIMES

Front Page | Taiwan News | Business | Editorials | Sports | World News | Fe

Home / Taiwan News

Sat, Dec 20, 2014 page4

Doctor shortage expected

MEDICAL WASTE: An E-Da Hospital doctor said that people blatantly waste medical resources, consuming a lot even though the nation is seeing birthrates decline

By Lin Hui-chin and Jake Chung / Staff reporter, with staff writer

The nation will likely face a shortage of doctors within eight years, a report by the National Health Research Institutes said yesterday.

Lack of Real-Time Monitoring

The New York Times

For Older Adults, Home Care Has Become Harder to Find

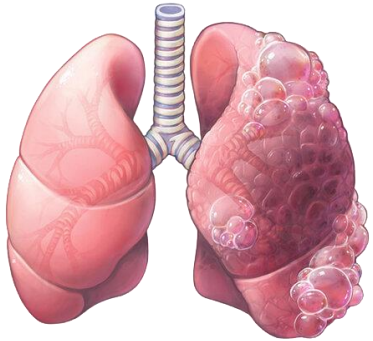
Staffing shortages have long plagued the home care industry. But the pandemic has intensified the problem.

Give this article | 143



The Importance of Real-Time Monitoring

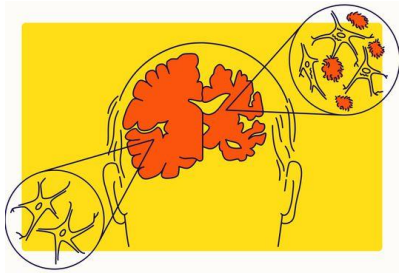
Vital Sign



Fall Detection



Sleep Pattern



Activity Recognition



Medical Complications

Daily Activity

ELDERLY STATISTICS

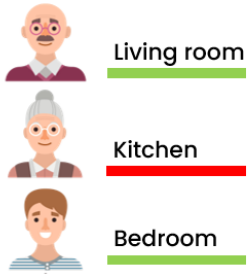


- 高齡人口占比趨勢 - 人口推估統計查詢系統. Ndc.gov.tw. Published 2022. https://pop-proj.ndc.gov.tw/chart.aspx?c=10&uid=66&pid=60
- iHealth. 台灣老人的健康狀況、住進機構意願、快樂比例 |. iHealth 健康報報. Published December 25, 2018. https://www.ihealth.com.tw/article/%E5%8F%B0%E7%81%A3%E8%80%81%E4%BA%BA%E5%BF%AB%E6%A8%82%E6%8C%87%E6%95%B8/



SOLUTION

Activity Recognition



Vital Sign



15BPM



LOOK AFTER

Non-intrusive
health monitoring device

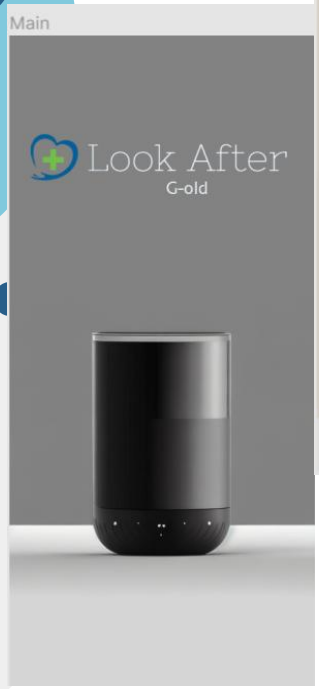
Fall Detection



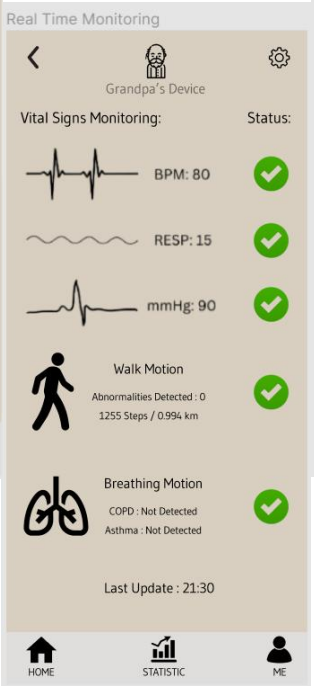
Sleep Monitoring

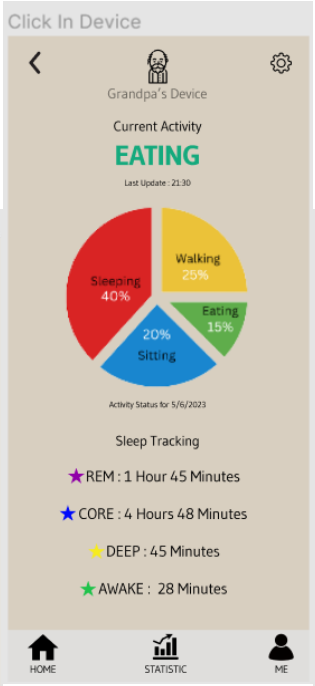


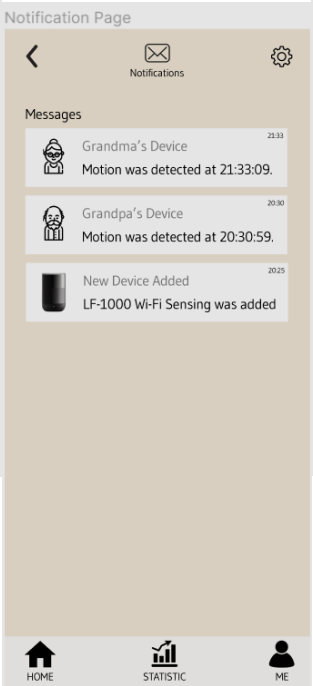
APP PROTOTYPE

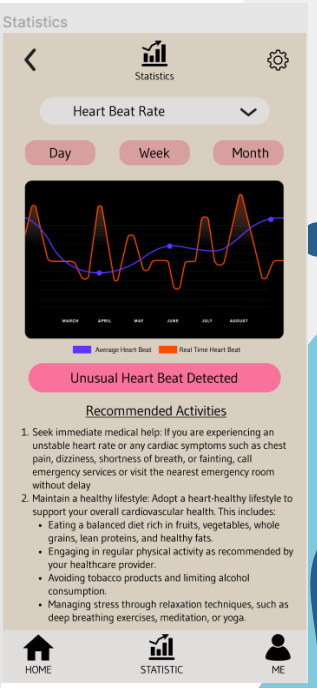












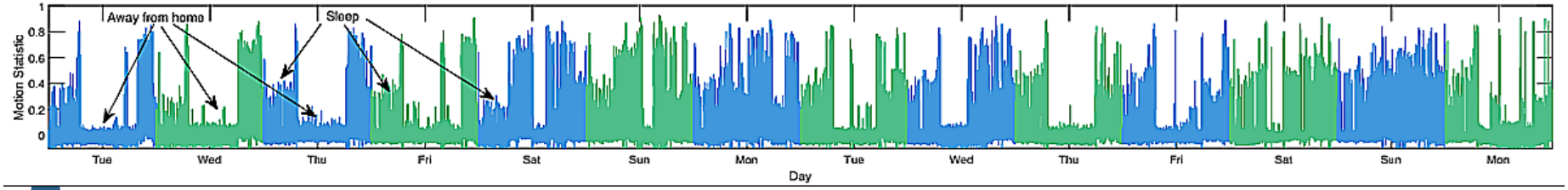
HOW IT WORKS

CSI (Channel State Information)

-Collect biosignal data

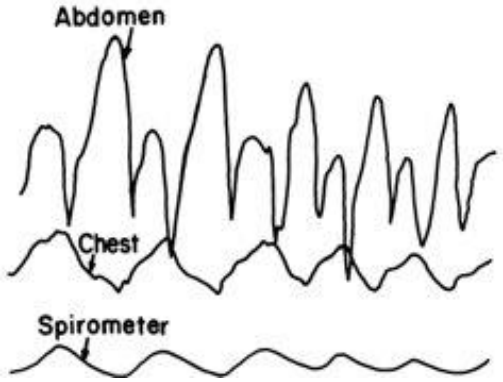
-Changes of Wi-Fi Signal



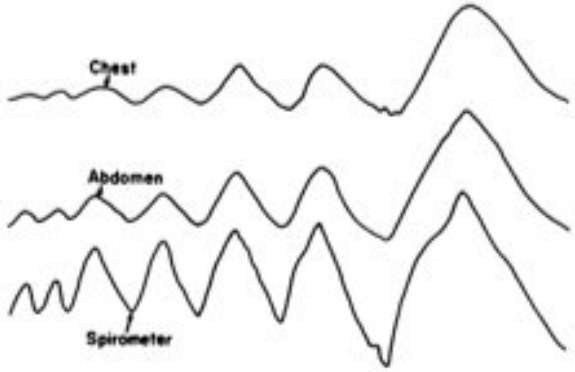


Motion Statistic

HOW IT WORKS



Abnormal breathing pattern



Normal breathing pattern

- Collect data
- Machine Learning
- Anomaly Detection
- Generate Insights

EXPECTED BENEFITS

- Faster -

Initiating immediate emergency response

- Cheaper -

Affordable health monitoring

- Better -

Delivering real-time health insights

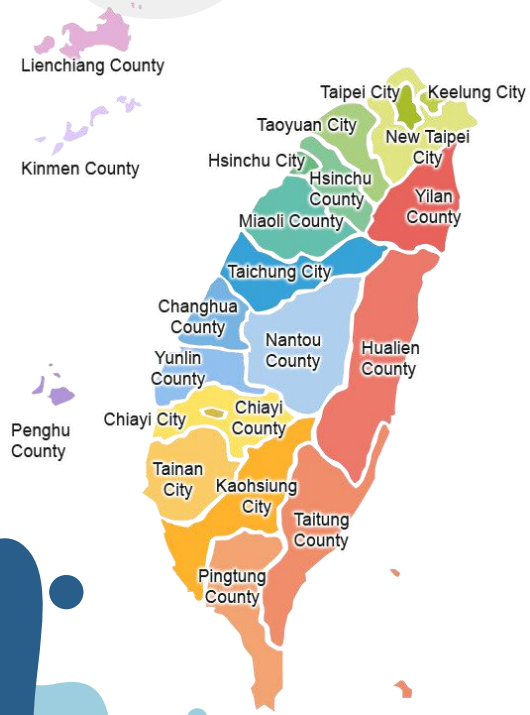
- Accessible -

Facilitation of remote patient monitoring

- Specialized -

Enabling proactive prevention through early detection

BEACH-HEAD MARKET



Tainan City

- The ratio of people over 65 accounted for **4 Million people**, around **18.03%** total population
- Accounted for 48.44% of assistance provided for the elder

Kaohsiung City

- The ratio of people over 65 accounted for **18.33%**
- Accounted for 44.06% assistance provided for the elder

BEACH-HEAD MARKET



Tainan City

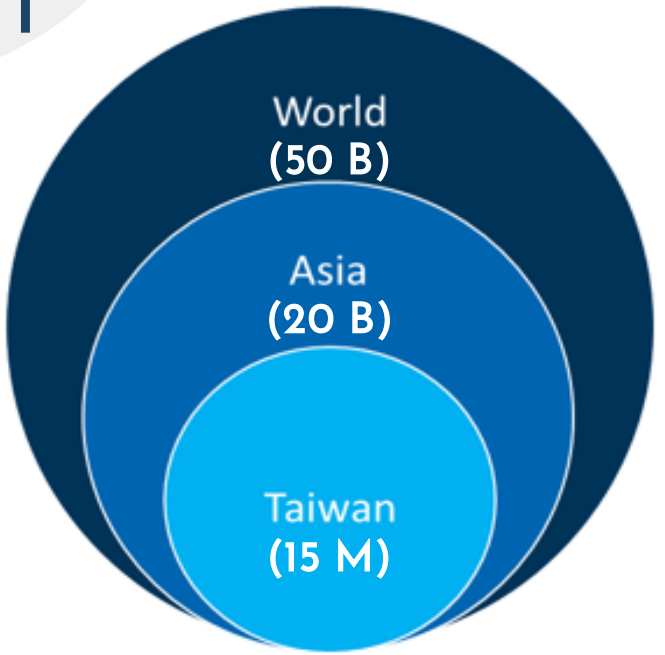
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Kaohsiung City

- The ratio of people over 65 accounted for **18.33%**
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FUTURE MARKET

In the year 2022, there are 313,399 applicants on Taiwan MOHW LTC 2.0 Program



313,399 x $50 = **$ 15,699,950**

(NTD 494,171,626)

COMPETITOR ANALYSIS

Product	Look After 	Google Nest Mini 	Apple Watch Series 8 	TP-Link C200 
Price	NT$1500=US$50	NT$1785 = USD$57	NT$12900 = USD$412	NT$849 = USD$27
User-friendly/Non-intrusive for elderlies	✓	✓		✓
Real-time vital sign monitoring	✓		✓	
Augmented analytics	✓			
Fall detection	✓		✓	

COST PER UNIT



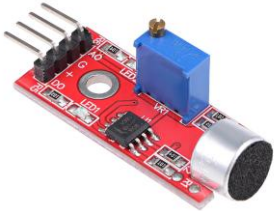
Product Shell (3D Printing)

$10



Grove Ultrasonic Ranger V2

$3



HW-484 (Sound Sensor)

$2



ESP32-CAM WIFI Bluetooth Board & OV2640 Camera Module with 2.4 Antenna CH340G

$12

Selling Price

$50

60

Cost

$30

PROJECT TIMELINE

Research of Wi-Fi Sensing Technology Commence



Pilot Test, Feedback Gather, Patient Data



Regulatory & Certification



Prototyping & Pre-Seed Funding USD 5,000



Product Design & Development



PROJECT TIMELINE

Application & Website Development, 2nd Funding USD 30,000



Project Research Done & Partners Seeking



Alpha & Beta Tests, Product Design & Project Development Done



3rd Funding USD 100,000



Launch & Mass Production



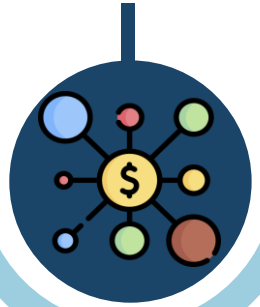
Customer Loyalty



PROJECT TIMELINE

Diversification





Introduce
New Features

Our story continues...

EXPECTED MARKET EXPAND

- **Stage 1: Taiwan**
  People over 65: 16.91% (2022)
- **Stage 2: Singapore**
  People over 65: 16.60% (2022)
- **Stage 3: Southeast Asia**
  People over 65: 9.50% (2022)



行政院
Executive Yuan

DKSH

RafflesMedicalGroup

MEDIHUB
BETTER HEALTH COMMUNITY

THOMSON MEDICAL

MGP
PT. MEGA GLOBAL PRATAMA
The Ultimate Trading Solution

TESTIMONIALS



Dr. Jun-Lian, Guo

Resident physician,
Department of Neurology,
Tri-service general hospital, Taipei

"It will make society become safer for those elderly population."



Dr. Ke-Wei, Chen

Attending physician,
Department of Family medicine,
Kaohsiung Armed Forces General Hospital

"This is a good idea, even in nursing homes, they could not monitor them continuously."



Dr. Cheok Kang Ni

General Doctor of Klinik
Kesihatan Tebrau, Johor, Malaysia

"It reduces the burden of caregivers as it does not need to look into the elderly all the time."

TEAM MEMBERS

G-OLD

NATALIE TANJAYA
Biomedical Engineering,
NCKU Taiwan



Chief Executive Officer (CEO)
Responsible for setting the overall direction and strategy.

LAM NGUYEN NGAN HA
Biomedical Engineering,
NCKU Taiwan



Chief Product Officer (CPO)
Overseeing the company's product or service development

YEOH ZHAO WEI
Software Engineering,
UTAR Malaysia



Chief Technology Officer (CTO)
Responsible for managing the company's finances

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

1931 國立成功大學 National Cheng Kung University

UTAR UNIVERSITI TUNKU ABDUL RAHMAN



SI POH YI
Civil Engineering, NTU
Singapore

Chief Marketing Officer (CMO)
Developing and executing the company's marketing strategy



CHAI JIN XIE
Civil Engineering,
NTU Singapore

Chief Tecnology Officer (CTO)
Develop and manage technology resources

Look After G-old

"Prevention is one of the few known ways to reduce demand for health and aged care" services."

Julie Bishop

(Australian Former Minister for Foreign Affairs, 1956)



Our Interview Video with Doctors

Thank you

SCAN ME

WE NEED YOU !!!



AI Experts
E&E Engineers
Legal Executives

Contact us:
C210058@e.ntu.edu.sg

Appendix

Financial Year 2024

	Project	Content	Cost Estimate (USD)
1	**Early Product Development**	**Wifi Sensor Design, AI System Establishment AND Wifi Sensor Settings with the Development Board**	**$2,000**
		Wifi Sensor Testing AND Prototype Design	**$3,000**
2	**Expert Consulting**	**Business Model, Regulations and Patent Experts**	**$7,000**
3	**Patent Layout**	**PA AND PCT**	**$15,000**
		Invention Patent, Utility Model Patent, Design Patent	**$900**
4	**Salary**	**Total Salary For 2024**	**$12,000**
5	**Operation in Taiwan**	**Business Registration**	**$38**
		Accounting and Audit (Every 14 Months)	**$1,400**
		Laboratory and Office Rental (Per Year)	**$25,000**
		Labour and Health Insurance (Per Year)	**$1000**
		Internet (HiNet) (Per Year)	**$600**
		Utility Bill (Per Year)	**$1,200**
6	**Regulatory Certification**	**(Taiwan) BSMI + NCC: certificate (per year)**	**$800**
7	**Others**	**Consumables and Transportation**	**$1,500**
Total			**$ 69,138** **NT$ 2,166,079.71**

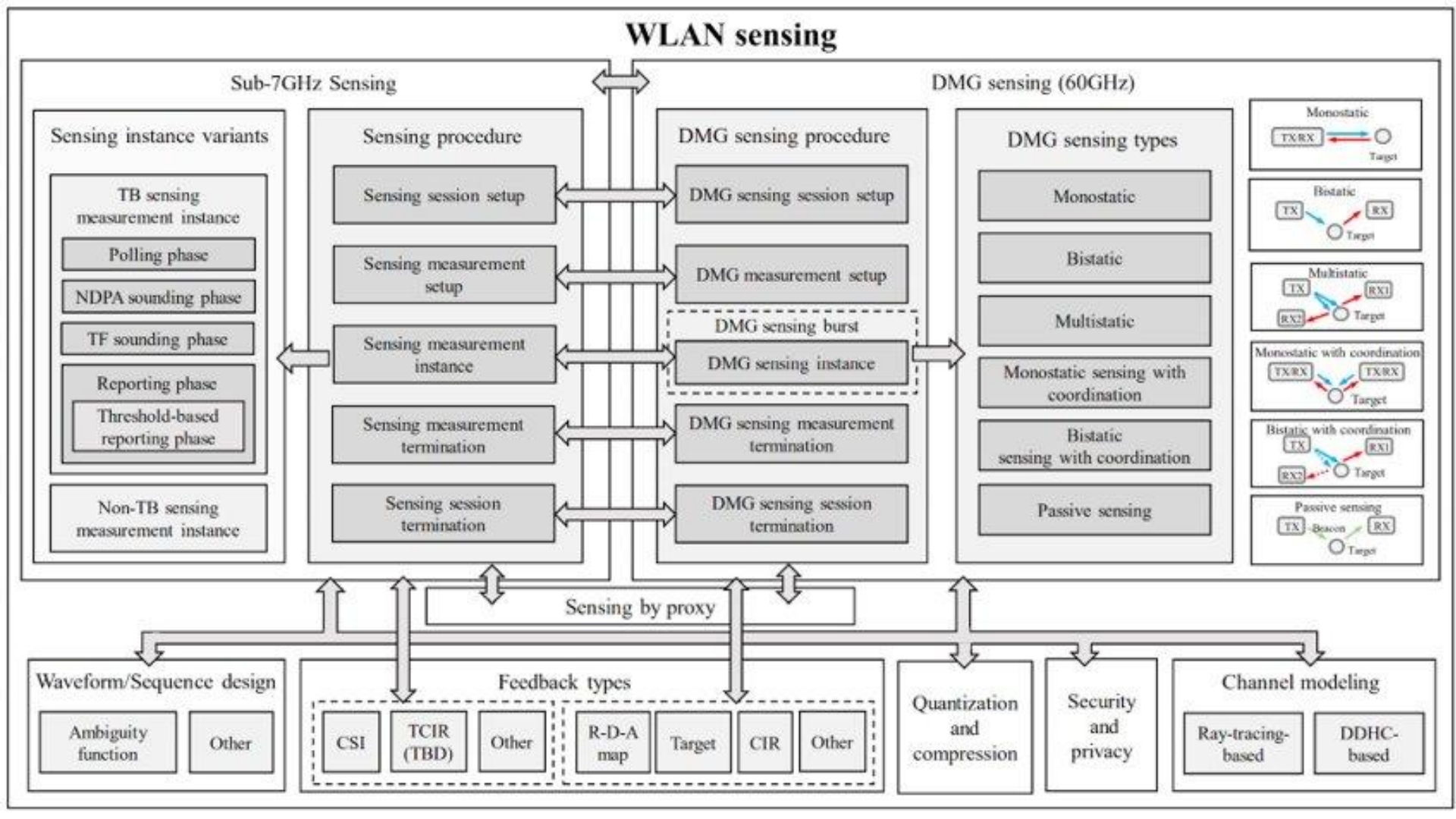
Financial Year 2025

	Project	Content	Cost Estimate (USD)
1	**Product Development**	**Wifi Sensor Alpha & Beta Testing**	**$5,000**
		Application Development AND Design	**$1,000**
		Adjustable Element Design AND Mass Production Test	**$5,000**
2	**Expert Consulting**	**Wifi Technology AND Device Testing Experts**	**$3,000**
3	**Salary**	**Total Salary For 2025**	**$60,000**
4	**Operation in Taiwan**	**Accounting and Audit (Every 14 Months)**	**$1,400**
		Laboratory and Office Rental (Per Year)	**$25,000**
		Labour and Health Insurance (Per Year)	**$1000**
		Internet (HiNet) (Per Year)	**$600**
		Utility Bill (Per Year)	**$1,200**
5	**Regulatory Certification**	**(Taiwan) BSMI + NCC: certificate (per year)**	**$800**
6	**Others**	**Consumables and Transportation**	**$3,500**
Total			**$ 107,500** **NT$ 3,367,953.50**

Financial Year 2026

	Project	Content	Cost Estimate (USD)
1	**Product Development**	**Mass Production**	**$20,000**
		Research AND Develop New Functions	**$5,000**
2	**Expert Consulting**	**Marketing Experts**	**$1,000**
4	**Salary**	**Total Salary For 2026**	**$90,000**
5	**Operation in Taiwan**	**Business Registration**	**$38**
		Accounting and Audit (Every 14 Months)	**$1,400**
		Laboratory and Office Rental (Per Year)	**$25,000**
		Labour and Health Insurance (Per Year)	**$1000**
		Internet (HiNet) (Per Year)	**$600**
		Utility Bill (Per Year)	**$1,200**
6	**Regulatory Certification**	**(Taiwan) BSMI + NCC: Laboratory test + certificate (per year), (Singapore) IMDA: Laboratory test + certificate (per year)**	**$900**
7	**Others**	**Consumables and Transportation**	**$3,500**
Total			**$ 146,138** **NT$ 4,578,474.31**

	2024	2025 - 2026
	Short-term	**Mid-term**
MILESTONES	**Marketing Site Launch** **BETA Launch** **Launch**	**Obtained 5% market in Taiwan – around 200,000 Users** **Approach international market (Singapore and Malaysia) with 100,000 Users**
PRODUCT	**Customer Testing** **Mobile App & Website** **Feedback**	**Customer care and after-sales service** **Usage Analytics** **System improvement** **Maintenance and update system**
SALES	**Create Bottom Up Start-up Strategy** **Analyze sales trends** **Competitor Analysis**	**Competitor Risk analysis (second phase)** **Connect with distribution networks**
MARKETING	**Lead Generation strategies** **Public Relations** **Set-up Google Analytics**	**Diversification** **Monthly webinar** **Introduce new features** **Loyalty campaign**
OPERATION	**Roles Assignment Matrix** **Create Expense Budget** **Functional Needs**	**Round funding raising** **Market sale reporting** **Expand company size**


WLAN sensing
Sub-7GHz Sensing
DMG sensing (60GHz)
Sensing instance variants
TB sensing measurement instance
Polling phase
NDPA sounding phase
TF sounding phase
Reporting phase
Threshold-based reporting phase
Non-TB sensing measurement instance
Sensing procedure
Sensing session setup
Sensing measurement setup
Sensing measurement instance
Sensing measurement termination
Sensing session termination
DMG sensing procedure
DMG sensing session setup
DMG measurement setup
DMG sensing burst
DMG sensing instance
DMG sensing measurement termination
DMG sensing session termination
DMG sensing types
Monostatic
Bistatic
Multistatic
Monostatic sensing with coordination
Bistatic sensing with coordination
Passive sensing
Monostatic
TX/RX
Target
Bistatic
TX
RX
Target
Multistatic
TX
RX1
RX2
Target
Monostatic with coordination
TX/RX
TX/RX
Target
Bistatic with coordination
TX
RX1
RX2
Target
Passive sensing
TX
Beacon
RX
Target
Sensing by proxy
Waveform/Sequence design
Ambiguity function
Other
Feedback types
CSI
TCIR (TBD)
Other
R-D-A map
Target
CIR
Other
Quantization and compression
Security and privacy
Channel modeling
Ray-tracing-based
DDHC-based

Key Partners

Healthcare Providers & Retailers
Technology Suppliers, Manufacturers, & Distributors
Government Agencies & Industry Players

Key Resources

WiFi Sensing Technology & Infrastructure
Funds
Data Storage & Analysis Capabilities
R&D Team
Partnerships & Collaborations
Regulatory Approvals
Brand Reputation & Customer's Loyalty
Customer Support Team

Customer Segments

Elderly with Respiratory Diseases
Caregivers of Elderly
Healthcare Institutions & Professionals
Smart Home Devices Enthusiasts or Early Adopters

Customer Relationships

Continuous Customer Support
Regular Software Updates & Improvements
Survey & Feedback Gathering
After-sales Support
Warranty
Loyalty Campaign

Key Activities

Short Term:

- Product Development
- Market Research
- Supply Chain & Partnerships
- Pre-launch Marketing & Promotion
- Public Relations

Mid Term:

- Expansion in Taiwan & Approach International Market
- Examine User's Usage & Improve Their Experience
- Expand Company Size & Tailor Future Roadmap
- Loyalty Campaign

Long Term:

- Expansion in other countries
- Introduce New Features
- Diversification
- Third Party Integration

Channels

Website, App & e-commerce platforms
Partnerships with healthcare institutions for distribution
Collaboration with smart home device manufacturers for bundled offerings
Healthcare seminars & conferences
Social media
Retail Partners

Key Value Propositions

Non-Intrusive Monitoring
Early Detection & Intervention
Real-time Personalized Care & Health Insights
User-friendly, Easy Installation & Affordable

Motivation

41% The age-adjusted fall death rate

Increased 55.3 per 100,000 older adults in 20121 to 78.0 per 100,000 older adults in 2021.

50% Sleep problems

The elderly complain about sleep problems compared to general generation at 20%

Vital sign Earlier abnormalities detection

Vital signs monitoring is crucial for living a long and healthy life

Activities Recognition Elderly are vulnerable group

Intervenes in case of change in behavior or a critical event has occurred

https://www.cdc.gov/falls/index.html

https://www.nia.nih.gov/health/good-nights-sleep#:

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3102151/

https://ieeexplore.ieee.org/document/9225639